

Cryptography & Security Lab

Formulas & Exam Reference Guide

CSE 4-1 Security Engineering

1 Modular Arithmetic & Number Theory

1. Modulo Operation:

$$r = a \bmod m = a - \left\lfloor \frac{a}{m} \right\rfloor \cdot m, \quad 0 \leq r < m$$

If negative: $r_{\text{pos}} = ((a \bmod m) + m) \bmod m$ [Test: $a = -23, m = 7 \implies r = 5$]

2. Euclidean GCD Algorithm:

$$\gcd(a, b) = \gcd(b, a \bmod b), \quad \gcd(a, 0) = a$$

Coprimality Condition: $\gcd(a, b) = 1$ [Test: $a = 60, b = 7 \implies \gcd(60, 7) = 1$]

3. Extended Euclidean Algorithm (Bézout's Identity):

$$a \cdot x + b \cdot y = \gcd(a, b)$$

Recursive Update:

$$x = y_1, \quad y = x_1 - \left\lfloor \frac{a}{b} \right\rfloor \cdot y_1$$

[Test: $a = 7, b = 60 \implies 7(43) + 60(-5) = 1 \implies x = 43, y = -5$]

4. Modular Multiplicative Inverse:

$$(a \cdot a^{-1}) \equiv 1 \pmod{m}$$

$$a^{-1} = (x \bmod m + m) \bmod m \quad (\text{where } a \cdot x + m \cdot y = 1)$$

[Test: $e = 7, \phi(n) = 60 \implies d = 7^{-1} \bmod 60 = 43$ (Check: $7 \times 43 = 301 \equiv 1 \bmod 60$)]

5. Fast Modular Exponentiation (Square-and-Multiply): To compute $b^e \bmod m$:

$$\text{While } e > 0 : \quad \text{if } e \bmod 2 == 1 \implies \text{res} = (\text{res} \cdot \text{base}) \bmod m$$

$$\text{base} = (\text{base}^2) \bmod m, \quad e = \lfloor e/2 \rfloor$$

[Test: $b = 5, e = 7, m = 77 \implies 5^7 \bmod 77 = 47$]

6. Euler's Totient Function & Theorem:

$$\boxed{\phi(p) = p - 1} \quad [\text{Test: } p = 7 \implies \phi(7) = 6]$$

$$\boxed{\phi(n) = (p - 1)(q - 1)} \quad [\text{Test: } p = 7, q = 11 \implies n = 77, \phi(n) = 6 \times 10 = \mathbf{60}]$$

$$\boxed{a^{\phi(n)} \equiv 1 \pmod{n} \quad (\text{if } \gcd(a, n) = 1)}$$

2 Classical Ciphers

1. Caesar Cipher:

- **Shift Key:** $k = 3$
- **Encryption Formula:**

$$\boxed{C = (P + k) \bmod 26}$$

$$[\text{Test: } P = \text{'H'}(7) \implies C = (7 + 3) \bmod 26 = 10(\text{'K'})]$$

- **Decryption Formula:**

$$\boxed{P = (C - k + 26) \bmod 26}$$

$$[\text{Test: } C = 10(\text{'K'}) \implies P = (10 - 3 + 26) \bmod 26 = 7(\text{'H'})]$$

2. Vernam Cipher (Stream / OTP):

- **Key Stream Condition:** $\text{length}(K) \geq \text{length}(P)$
- **Additive Stream:**

$$\boxed{C_i = (P_i + K_i) \bmod 26, \quad P_i = (C_i - K_i + 26) \bmod 26}$$

$$[\text{Test: } P_i = 4(\text{'E'}), K_i = 9(\text{'J'}) \implies C_i = 13(\text{'N'})]$$

- **Bitwise XOR (OTP):**

$$\boxed{C_i = P_i \oplus K_i, \quad P_i = C_i \oplus K_i}$$

$$[\text{Test: } P_i = 1010_2, K_i = 1100_2 \implies C_i = 0110_2 \implies P_i = 0110_2 \oplus 1100_2 = 1010_2]$$

3 RSA Cryptosystem

1. Setup Parameters to Take in Exam:

- **Primes:** $p = 7, \quad q = 11$
- **Modulus:** $n = p \cdot q = 7 \times 11 = 77$
- **Totient:** $\phi(n) = (7 - 1)(11 - 1) = 6 \times 10 = 60$
- **Public Exponent:** $e = 7 \quad (\gcd(7, 60) = 1)$
- **Private Exponent:** $d = 7^{-1} \bmod 60 = 43 \quad (7 \times 43 = 301 \equiv 1 \bmod 60)$
- **Messages:** $m_1 = 5, \quad m_2 = 6$

2. Key Pair:

$$\text{Public Key: } (e, n) = (7, 77)$$

$$\text{Private Key: } (d, n) = (43, 77)$$

3. Encryption & Decryption:

$$c = m^e \bmod n$$

$$[\text{Test: } c_1 = 5^7 \bmod 77 = 47, \quad c_2 = 6^7 \bmod 77 = 41]$$

$$m = c^d \bmod n$$

$$[\text{Test: } m_1 = 47^{43} \bmod 77 = 5]$$

4. Digital Signature:

$$S = m^d \bmod n \quad [\text{Test: } S = 5^{43} \bmod 77 = 26]$$

$$m' = S^e \bmod n \quad [\text{Test: } m' = 26^7 \bmod 77 = 5 == m \implies \text{Valid Signature}]$$

5. Multiplicative Homomorphism:

$$c_{\text{comb}} = (c_1 \cdot c_2) \bmod n$$

$$[\text{Test: } c_{\text{comb}} = (47 \times 41) \bmod 77 = 1927 \bmod 77 = 2]$$

$$M = (c_{\text{comb}})^d \bmod n = (m_1 \cdot m_2) \bmod n$$

$$[\text{Test: } M = 2^{43} \bmod 77 = 30 = (5 \times 6) \bmod 77]$$

4 ElGamal Cryptosystem

1. Setup Parameters to Take in Exam:

- **Prime:** $p = 467$
- **Generator:** $\alpha = 2$
- **Private Key:** $a = 127$
- **Public Parameter:** $\beta = \alpha^a \bmod p = 2^{127} \bmod 467 = 132$
- **Messages:** $M_1 = 10, \quad M_2 = 20$
- **Random Keys:** $k_1 = 5, \quad k_2 = 7$

2. Key Pair:

$$\text{Public Key: } (p, \alpha, \beta) = (467, 2, 132)$$

$$\text{Private Key: } a = 127$$

3. Standard Encryption & Decryption:

$$c_1 = \alpha^k \bmod p, \quad c_2 = (M \cdot \beta^k) \bmod p$$

$$[\text{Test: } c_{11} = 2^5 \bmod 467 = \mathbf{32}, \quad c_{12} = (10 \times 132^5) \bmod 467 = \mathbf{454} \implies C_1 = (32, 454)]$$

$$[s = c_1^a \bmod p = \beta^k \bmod p \quad [\text{Test: } s = 32^{127} \bmod 467 = \mathbf{326}, \quad s^{-1} \bmod 467 = \mathbf{239}]]$$

$$M = (c_2 \cdot s^{-1}) \bmod p = (c_2 \cdot (c_1^a)^{-1}) \bmod p$$

$$[\text{Test: } M = (454 \times 239) \bmod 467 = \mathbf{10}]$$

4. Multiplicative Homomorphism:

$$c_{\text{comb1}} = (c_{11} \cdot c_{21}) \bmod p, \quad c_{\text{comb2}} = (c_{12} \cdot c_{22}) \bmod p$$

$$[\text{Test: } C_2 = \text{Encrypt}(M_2 = 20, k_2 = 7) = (128, 393)]$$

$$[\text{Test: } c_{\text{comb}} = ((32 \times 128) \% 467, (454 \times 393) \% 467) = (\mathbf{351}, \mathbf{375})]$$

$$\text{Decrypt}(c_{\text{comb1}}, c_{\text{comb2}}) = (M_1 \cdot M_2) \bmod p$$

$$[\text{Test: } \text{Decrypt}(351, 375) = \mathbf{200} = (10 \times 20) \bmod 467]$$

5. Re-Randomization (Random $r = 3$):

$$c'_1 = (c_1 \cdot \alpha^r) \bmod p, \quad c'_2 = (c_2 \cdot \beta^r) \bmod p$$

$$[\text{Test: } \text{Take } r = 3 \implies \alpha^3 = 8, \quad \beta^3 = 132^3 \bmod 467 = 447]$$

$$[\text{Test: } c'_1 = (32 \times 8) \bmod 467 = \mathbf{256}, \quad c'_2 = (454 \times 447) \bmod 467 = \mathbf{274}]$$

$$[\text{Test: } \text{Decrypt}(256, 274) = \mathbf{10} = M]$$

6. Exponential ElGamal (Additive):

$$c_1 = \alpha^k \bmod p, \quad c_2 = (\alpha^M \cdot \beta^k) \bmod p$$

$$(c_{11} \cdot c_{21}, c_{12} \cdot c_{22}) \implies (\alpha^{k_1+k_2}, \alpha^{M_1+M_2} \cdot \beta^{k_1+k_2})$$

7. ElGamal Digital Signature:

- **Setup:** Message $M = 100$, Ephemeral Random $r = 5 \implies r^{-1} \bmod 466 = 373$
- **Signing Formulas:**

$$y_1 = \alpha^r \bmod p \quad [\text{Test: } y_1 = 2^5 \bmod 467 = 32]$$

$$y_2 = [r^{-1} \cdot (M - a \cdot y_1)] \bmod (p - 1)$$

$$[\text{Test: } y_2 = [373(100 - 127 \times 32)] \bmod 466 = 82 \implies \text{Sig: } (32, 82)]$$

- **Verification Formula:**

$$\alpha^M \equiv (\beta^{y_1} \cdot y_1^{y_2}) \pmod{p}$$

$$[\text{Test: } 2^{100} \bmod 467 = 78 \equiv (132^{32} \times 32^{82}) \bmod 467 = 78 \implies \text{Valid Signature}]$$

5 Elliptic Curve Cryptography (ECC) & ECDSA

1. Setup Parameters to Take in Exam:

- **Prime Field:** $p = 17$
- **Curve Equation:** $a = 2, b = 2 \implies y^2 \equiv x^3 + 2x + 2 \pmod{17}$
- **Non-Singularity:** $4a^3 + 27b^2 = 4(8) + 27(4) = 140 \equiv 4 \not\equiv 0 \pmod{17}$
- **Base Point:** $G = (5, 1) \quad (1^2 \equiv 5^3 + 2(5) + 2 = 137 \equiv 1 \pmod{17})$
- **Order of G :** $n = 19$
- **Private Key:** $d = 2$
- **Public Key Point:** $Q = d \cdot G = 2(5, 1) = (6, 3)$

2. Key Pair:

$$\text{Public Key: } Q = (6, 3) \quad (\text{with } p = 17, a = 2, b = 2, G = (5, 1), n = 19)$$

$$\text{Private Key: } d = 2$$

3. Point Identity & Inverse:

$$\mathcal{O} = \text{Point at infinity} \quad (P + \mathcal{O} = P)$$

$$-P = (x_1, (-y_1 \bmod p + p) \bmod p) \quad (P + (-P) = \mathcal{O})$$

4. Point Doubling & Addition ($R = P + Q = (x_3, y_3)$):

- **Point Doubling** ($P = Q = (5, 1)$):

$$\lambda = [(3x_1^2 + a) \cdot (2y_1)^{-1}] \bmod p$$

$$[\text{Test: } \lambda = [(3 \cdot 25 + 2) \cdot (2 \cdot 1)^{-1}] \bmod 17 = 77 \times 9 \equiv 9 \times 9 = 81 \equiv \mathbf{13}]$$

$$x_3 = (\lambda^2 - 2x_1) \bmod p, \quad y_3 = [\lambda(x_1 - x_3) - y_1] \bmod p$$

$$[\text{Test: } x_3 = (13^2 - 10) \bmod 17 = \mathbf{6}, \quad y_3 = [13(5 - 6) - 1] \bmod 17 = \mathbf{3} \implies 2G = (\mathbf{6}, \mathbf{3})]$$

- **Point Addition** ($P(5, 1) \neq Q(6, 3)$):

$$\lambda = [(y_2 - y_1) \cdot (x_2 - x_1)^{-1}] \bmod p$$

$$[\text{Test: } \lambda = [(3 - 1)(6 - 5)^{-1}] \bmod 17 = \mathbf{2}]$$

$$x_3 = (\lambda^2 - x_1 - x_2) \bmod p, \quad y_3 = [\lambda(x_1 - x_3) - y_1] \bmod p$$

$$[\text{Test: } x_3 = (2^2 - 5 - 6) \bmod 17 = \mathbf{10}, \quad y_3 = [2(5 - 10) - 1] \bmod 17 = \mathbf{6} \implies 3G = (\mathbf{10}, \mathbf{6})]$$

5. Scalar Multiplication:

$$kP = \underbrace{P + P + \dots + P}_{k \text{ times}} \quad \text{via Double-and-Add Algorithm}$$

6. ECC ElGamal Public-Key Encryption:

- **Message Point:** $P_m = (10, 6)$, **Random:** $k = 2$

- **Encryption Formulas:**

$$C_1 = k \cdot G, \quad C_2 = P_m + k \cdot Q$$

$$[\text{Test: } C_1 = 2(5, 1) = (\mathbf{6}, \mathbf{3}), \quad C_2 = (10, 6) + 2(6, 3) = (10, 6) + (3, 1) = (\mathbf{0}, \mathbf{6})]$$

$$[\text{Ciphertext: } C = ((\mathbf{6}, \mathbf{3}), (\mathbf{0}, \mathbf{6}))]$$

- **Decryption Formula:**

$$P_m = C_2 - d \cdot C_1 = C_2 + (-d \cdot C_1)$$

$$[\text{Test: } P_m = (0, 6) - 2(6, 3) = (0, 6) - (3, 1) = (0, 6) + (3, 16) = (\mathbf{10}, \mathbf{6}) = P_m]$$

7. ECC Homomorphism (Additive):

$$P_1 = m_1 G, \quad P_2 = m_2 G \implies P_{\text{sum}} = P_1 + P_2 = (m_1 + m_2)G$$

$$[\text{Test: } m_1 = 2 \implies P_1 = (6, 3), \quad m_2 = 1 \implies P_2 = (5, 1) \implies P_{\text{sum}} = (6, 3) + (5, 1) = (\mathbf{10}, \mathbf{6}) = 3G]$$

8. ECDSA (Elliptic Curve Digital Signature Algorithm):

- **Setup:** Message $M = 7$, Private key $d = 2$, Random $k = 3$
- **Signature Generation Formulas:**

$$\boxed{R = (x_R, y_R) = k \cdot G} \quad [\textbf{Test: } R = 3(5, 1) = (\mathbf{10}, \mathbf{6})]$$

$$\boxed{r = x_R \bmod n} \quad [\textbf{Test: } r = 10 \bmod 19 = \mathbf{10}]$$

$$\boxed{s = [k^{-1} \cdot (M + d \cdot r)] \bmod n}$$

$$[\textbf{Test: } 3^{-1} \bmod 19 = 13 \implies s = [13(7 + 2 \times 10)] \bmod 19 = (13 \times 27) \bmod 19 = \mathbf{9}]$$

$$[\textbf{Signature Pair: } (r, s) = (\mathbf{10}, \mathbf{9})]$$

- **Signature Verification Formulas:**

$$\boxed{w = s^{-1} \bmod n} \quad [\textbf{Test: } w = 9^{-1} \bmod 19 = \mathbf{17}]$$

$$\boxed{u_1 = (M \cdot w) \bmod n, \quad u_2 = (r \cdot w) \bmod n}$$

$$[\textbf{Test: } u_1 = (7 \times 17) \bmod 19 = \mathbf{5}, \quad u_2 = (10 \times 17) \bmod 19 = \mathbf{18}]$$

$$\boxed{X = (x_X, y_X) = u_1 \cdot G + u_2 \cdot Q}$$

$$[\textbf{Test: } X = 5(5, 1) + 18(6, 3) = (\mathbf{10}, \mathbf{6})]$$

$$\boxed{\text{Valid Signature if and only if: } v = x_X \bmod n == r}$$

$$[\textbf{Test: } v = 10 \bmod 19 = \mathbf{10} == r \implies \textbf{SIGNATURE VALID}]$$